

MARKOV'S INEQUALITY

$\Rightarrow$ If $\bar{X} \geq 0$

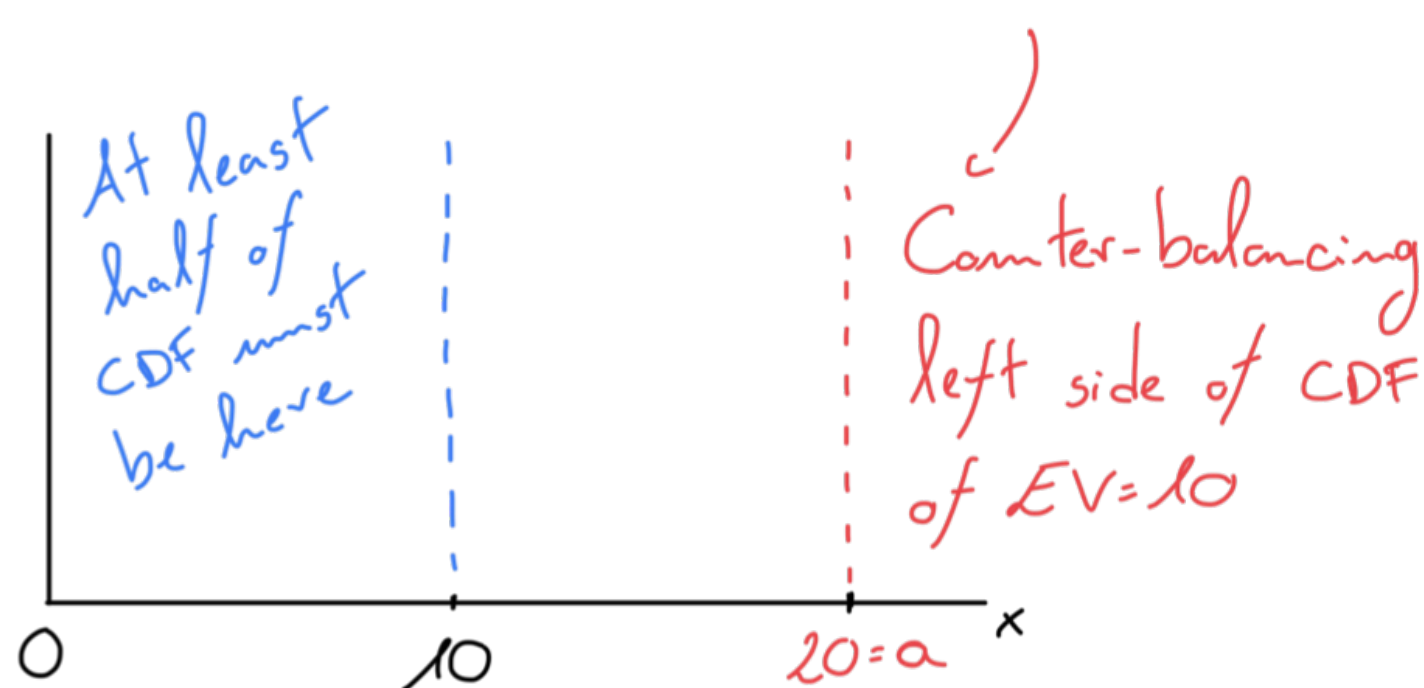
$$\text{then } P(\bar{X} \geq a) \leq \frac{E(\bar{X})}{a}$$

$$\begin{aligned} E(\bar{X}) &= \sum_x x P(\bar{X}=x) \geq \sum_{x \geq a} x P(\bar{X}=x) \\ &\geq \sum_{x \geq a} a P(\bar{X}=x) \\ &= \underline{a P(\bar{X} \geq a)} \end{aligned}$$

GRAPHICAL EXAMPLE

$$E(\bar{X}) = 10$$

$$P(\bar{X} \geq 20) \leq 1/2$$



In poor terms: You can't have too much mass of the distribution too far away from the μ .

CHEBYSHEV INEQUALITY

$$E(\bar{X}) = \mu$$

$$\text{VAR}(\bar{X}) = \sigma^2$$

$\Rightarrow$ For any $a > 0$ then

$$P(|\bar{X} - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Proof: $\bar{Y} = (\bar{X} - \mu)^2$ $b = a^2$

$$\begin{aligned} P(|\bar{X} - \mu| \geq a) &= P((\bar{X} - \mu)^2 \geq a^2) \\ &= P(\bar{Y} \geq b) \end{aligned}$$

Using $E(\bar{Y}) = E((\bar{X} - \mu)^2) = \text{VAR}(\bar{X}) = \sigma^2$
& MARKOV'S INEQUALITY

$$P(\bar{Y} \geq b) \leq \frac{E(\bar{Y})}{b} \leq \frac{\sigma^2}{a^2}$$

In poor terms: If I have $\text{VAR}(\bar{X}) = \sigma^2$ I can't have too many of my points $\bar{X}$ spreading away from μ . The prob. of having $\bar{X}$ far away of a from μ is bound to σ^2 .